

Hydrochloric Acid 33% — Aqueous Solution

PRODUCT DATA SHEET
FT-HCL33-2024

1. PRODUCT IDENTIFICATION

CAS No.	7647-01-0	EC No.	231-595-7
Formula	HCl (aq.) — 33% w/w	Molar mass	36.46 g/mol (pure HCl)
Chemical family	Strong mineral acid	Grade	Technical

Available pack sizes: 25 L jerrican (HDPE) | 200 L drum (HDPE) | 1000 L IBC (HDPE)

2. PHYSICO-CHEMICAL PROPERTIES

Property	Value
Physical state	Colourless to pale yellow liquid
Odour	Pungent, acrid (HCl fumes)
Boiling point (azeotrope)	108.6 °C at 20.2% HCl
Freezing point	-36 °C (HCl 33%)
Density at 20°C	1.155–1.165 g/mL
Viscosity at 20°C	1.9 mPa·s
HCl vapour pressure at 20°C	~16 hPa
pH (undiluted)	< 0
Electrical conductivity	High (strong electrolyte)

3. ANALYTICAL SPECIFICATIONS

Parameter	Specification
HCl content	32.0–34.0%
Density at 20°C	1.155–1.165 g/mL
Evaporation residue	<= 0.01%
Sulfates (SO4 ²⁻)	<= 5 ppm
Iron (Fe)	<= 3 ppm
Arsenic (As)	<= 0.1 ppm
Heavy metals (Pb)	<= 1 ppm
Colour (APHA)	<= 20

4. APPLICATIONS AND USES

- Acid pickling of steel before galvanising or welding
- Regeneration of cationic ion-exchange resins
- Synthesis of metal chlorides (FeCl₂, ZnCl₂, AlCl₃)
- pH adjustment in water treatment
- Gelatine extraction and protein hydrolysis
- Cleaning of tanks and pipework in the food industry

5. HAZARD INFORMATION (CLP/GHS)

Code	Hazard statement
H290	May be corrosive to metals
H314	Causes severe skin burns and eye damage
H335	May cause respiratory irritation

6. PRECAUTIONARY STATEMENTS

Code	Precautionary statement
P260	Do not breathe vapours
P271	Use only outdoors or in a well-ventilated area
P280	Acid-resistant gloves, eye and face protection
P301+P330+P331	IF SWALLOWED: rinse mouth. Do NOT induce vomiting
P304+P340	IF INHALED: move the person to fresh air

7. STORAGE AND SHELF LIFE

Store in acid-resistant HDPE or glass containers in a ventilated area away from bases, oxidisers and metals. Storage temperature: 0–30°C. Shelf life: 24 months (risk of off-gassing at elevated temperature).

8. TRANSPORT AND REGULATIONS

UN 1789 | ADR class 8 | Packing group II | Tunnel code E

9. DETAILED APPLICATION NOTES

Material compatibility

Before use at scale, check compatibility with seals, pipework and containers. Give preference to materials tested and validated by ChemCorp Industries. Refer to the chemical compatibility tables available on our technical portal (portail.chemcorp.fr/compat). If in doubt, contact our Technical Application service on +33 4 72 XX XX XX.

Quality control on receipt

On receipt, check the integrity of the packaging and that the batch numbers match the accompanying certificate of analysis. Carry out a density and pH check where available. In the event of an anomaly, keep the original packaging and contact our quality department within 48 hours. Procedure ref.: QC-RECEP-003.

Waste management

Waste from this product must be managed in accordance with local hazardous waste regulations (Directive 2008/98/EC). Contact an approved contractor for disposal. European Waste Code (EWC): 06 01 04* for acid solutions, 07 01 01* for halogenated solvents, 07 01 04* for other organic solvents. Never discharge into drains or watercourses.

First aid (operational summary)

INHALATION: Move the person to fresh air, in a semi-seated position. If breathing is difficult, administer O2 and call the emergency services (15 in France). SKIN CONTACT: Remove contaminated clothing. Rinse with plenty of water for at least 15 minutes. EYE CONTACT: Rinse with water for at least 15 minutes, holding the eyelids open. Consult an ophthalmologist. INGESTION: Do not induce vomiting. Rinse the mouth. Call the Poison Control Centre (European number: 0800 59 59 59).

Environmental data (in-house results)

BOD5 (5 days): tested per ISO 5815. COD: tested per ISO 15705. Log Kow (n-octanol/water partition coefficient): measured per OECD 117. Aerobic biodegradability: tested per OECD 301B. Aquatic toxicity (Daphnia magna): EC50 per OECD 202. For the complete numerical values, refer to section 12 of the Safety Data Sheet.

Additional regulatory information

This product is registered under REACH: registration number available on request. It is / is not on the SVHC candidate list (last ECHA update: Jan 2025). IED Directive: applicable to installations using this product in quantities above 200 t/year. French ICPE: check the classification under the ICPE nomenclature (heading 1175, 4110 or 4XXX depending on the product). Certifications: ISO 9001:2015 | Compliant with IED Directive 2010/75/EU.